

Compute Core — Mathematical Description

5-Class Cardiac Arrhythmia Classifier, RBF-SVM ASIC

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The `svm_compute_core` implements two sequential subfunctions per support vector: a **distance function** (`dist`) that computes the squared Euclidean distance between an input feature vector and a support vector, and a **kernel function** (`horner`) that evaluates the RBF exponential. After all support vectors are processed, an **accumulation and classification** step produces the output class label. All arithmetic is performed in $\mathbb{Q}_{6.10}$ 16-bit signed fixed-point unless otherwise noted.

Fixed-Point Representation ($\mathbb{Q}_{6.10}$)

A 16-bit signed value v is stored as the integer

$$\hat{v} = \lfloor v \cdot 2^{10} \rfloor \in [-32768, 32767]$$

with the correspondence

$$v = \frac{\hat{v}}{1024}, \quad \text{range: } [-32, +31.999], \quad \text{LSB} = 2^{-10} \approx 0.001.$$

Multiplication of two $\mathbb{Q}_{6.10}$ values $\hat{a}$ and $\hat{b}$ produces a 32-bit intermediate result that is right-shifted by 10 bits to restore the $\mathbb{Q}_{6.10}$ format:

$$\widehat{a \cdot b} = \frac{\hat{a} \cdot \hat{b}}{2^{10}}.$$

The parameter $\gamma = 0.25$ is exactly representable: $\hat{\gamma} = 0.25 \times 1024 = 256 = 0x0100$.

Distance Function (`dist`)

Definition

For input feature vector $\mathbf{x} \in \mathbb{Q}_{6.10}^N$ and support vector $\mathbf{sv} \in \mathbb{Q}_{6.10}^N$ with $N = 256$ dimensions:

$$d^2(\mathbf{x}, \mathbf{sv}) = \sum_{i=0}^{N-1} (x_i - sv_i)^2.$$

Pipeline Implementation

The accumulator is a two-stage registered pipeline:

$$\delta_i = x_i - sv_i, \quad s_i = \delta_i^2, \quad D_k = \sum_{i=0}^k s_i.$$

Stage 1 (cycle c): compute and register $s_i = \delta_i^2$.

Stage 2 (cycle $c + 1$): add s_i to the running accumulator D .

After the N -th input pair is presented, **two drain cycles** are required before the result D_{N-1} is valid:

- Drain cycle 1: s_{N-1} propagates from the multiplier register to the adder input.
- Drain cycle 2: s_{N-1} is added into D and the result is registered.

Total cycles per SV distance: $N + 2 = 258$ at $\text{LAT} = 1$; $N \cdot \text{LAT} + 2$ at general latency.

Saturation

The accumulator is 32-bit, with a hardware maximum of $2^{32} - 1$. The theoretical worst-case accumulated value for $N = 256$ and maximum feature magnitude $|x_i - sv_i| \leq 63.999$ is

$$D_{\max} = 256 \times (63.999)^2 \approx 1,048,544 \approx 2^{20},$$

which is well within the 32-bit range. The fixed-point overflow boundary is determined by the gamma multiply: after right-shifting by 2^{10} , the maximum representable $\mathbb{Q}_{6.10}$ value from a 32-bit accumulator is

$$\frac{2^{32}}{2^{10}} = 2^{22}.$$

With $\hat{\gamma} = 2^8$, the accumulator would produce $\hat{p} > 2^{18}$ only if $D > 2^{20}$, which does not occur on real ECG data. The `ERR_DIST_OVERFLOW` flag is therefore a defensive sanity check rather than a condition expected in normal operation.

Kernel Function (horner)

Definition

The RBF (Gaussian) kernel is

$$K(\mathbf{x}, \mathbf{sv}) = e^{-\gamma d^2(\mathbf{x}, \mathbf{sv})}, \quad \gamma = 0.25.$$

The argument $p = \gamma d^2$ ranges over $[0, \gamma D_{\max}] \approx [0, 262,136/1024] \approx [0, 256]$ in floating-point. In $\mathbb{Q}_{6.10}$ fixed-point, $\hat{\gamma} = 256 = 2^8$ and $\hat{d}^2$ reaches up to $\sim 2^{20}$ (saturating accumulator), so after the $\mathbb{Q}_{6.10}$ multiply (right-shift by 2^{10}):

$$\hat{p} = \frac{\hat{\gamma} \cdot \hat{d}^2}{2^{10}} \leq \frac{2^8 \cdot 2^{20}}{2^{10}} = 2^{18}.$$

Range Reduction

Direct Horner evaluation of e^{-p} over $[0, 256]$ requires degree ≥ 8 and overflows $\mathbb{Q}_{6.10}$ intermediate products. Instead, p is split into integer and fractional parts:

$$I = \left\lfloor \frac{\hat{p}}{2^{10}} \right\rfloor \in [0, 255], \quad F = \frac{\hat{p} \bmod 2^{10}}{2^{10}} \in [0, 1),$$

so that

$$e^{-p} = e^{-I} \cdot e^{-F}.$$

LUT term: e^{-I}

A 256-entry read-only table stores

$$\text{lut}[I] = \lfloor e^{-I} \cdot 2^{10} \rfloor \in \mathbb{Q}_{6.10}, \quad I = 0, 1, \dots, 255.$$

For $I \geq 8$, $e^{-I} < 0.00034$, which rounds to zero in $\mathbb{Q}_{6.10}$, so the effective range is $I \in [0, 7]$. The LUT occupies 256×16 bits = 4 KB of on-chip flip-flop registers.

Polynomial term: e^{-F}

For $F \in [0, 1)$, numerical analysis shows that degree 6 is the minimum Taylor polynomial that keeps the approximation error below the $\mathbb{Q}_{6.10}$ LSB of $2^{-10} \approx 0.001$ across the full range. The RTL defines up to 16 Taylor coefficients (`COEFF_00`–`COEFF_15`) and evaluates them via Horner’s method:

$$e^{-F} \approx \sum_{n=0}^d \frac{(-F)^n}{n!} = c_0 + F(c_1 + F(c_2 + \dots + F \cdot c_d)),$$

where the coefficients are the Taylor series values $c_n = 1/n!$ with alternating signs applied during evaluation. Evaluation proceeds right-to-left (Horner’s method), requiring d multiplications and d additions.

Kernel Output

The two terms are combined by a $\mathbb{Q}_{6.10}$ multiply:

$$\hat{K} = \frac{\text{lut}[I] \cdot \hat{e}^{-F}}{2^{10}} \in [0, 1024],$$

where $\hat{K} = 1024$ corresponds to $K = 1.0$ (achieved when $d^2 = 0$, i.e. $\mathbf{x} = \mathbf{sv}$).

Alpha Accumulation and OVR Decision

Per-Class Accumulation

For each support vector $j = 0, \dots, M - 1$ with class label $\ell_j \in \{0, 1, 2, 3, 4\}$ and alpha coefficient $\alpha_j \in \mathbb{Q}_{6.10}$, the kernel output is weighted and accumulated into the corresponding class score:

$$f_c(\mathbf{x}) = \sum_{\substack{j=0 \\ \ell_j=c}}^{M-1} \alpha_j \cdot K(\mathbf{x}, \mathbf{sv}_j), \quad c = 0, 1, 2, 3, 4.$$

The five accumulators $f_0, \dots, f_4$ are 32-bit registers, updated after each SV kernel evaluation. Alpha coefficients are stored on-chip in a 500-entry register file (`alpha_table[]`), loaded via the Wishbone `ALPHA_WR` register before classification begins.

Classification (Argmax)

After all $M = 500$ support vectors are processed, the predicted class is

$$\hat{y} = \arg \max_{c \in \{0, 1, 2, 3, 4\}} f_c(\mathbf{x}).$$

The argmax is implemented as a 5-way comparator tree and completes in one clock cycle. The result is registered on `class_out[2:0]` and exposed on GPIO.

Operation Count Summary

Stage	Operations per SV	SVs	Total
dist: subtractions	256	500	128,000
dist: multiplications	256	500	128,000
dist: accumulations	256	500	128,000
horner: multiplications	3	500	1,500
horner: additions	3	500	1,500
Alpha multiply-accumulate	1	500	500
Total per beat			387,500

At $N = 256$ and $M = 500$, the design is strongly **memory-bound**: arithmetic intensity $AI \approx 1.5$ OPs/byte (LUT preloaded), placing the operating point well below the compute ceiling of the 40 MHz fixed-point datapath.

FSM Cycle Budget (LAT = 3)

$$\begin{aligned} T_{\text{beat}} &= T_{\text{load}} + M \cdot (T_{\text{dist}} + T_{\text{kernel}}) + T_{\text{classify}} \\ &= N \cdot \text{LAT} + M (N \cdot \text{LAT} + 2 + 18) + 1 \\ &= 256 \times 3 + 500 \times (256 \times 3 + 20) + 1 = 768 + 500 \times 788 + 1 = 394,769 \text{ cycles/beat.} \end{aligned}$$

At 40 MHz: $394,769/40 \times 10^6 \approx 9.87$ ms/beat, consistent with the measured 9.7 ms (the small difference reflects pipeline overlap between LOAD and COMPUTE).